Alberto Trevisan

PhD student · Department of Engineering and Architecture, University of Trieste

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Research interests

Coordination in multi-agent systems through reinforcement learning, variance-based assessment of safety, credit assignment and partial observability, decentralized formation control of drone swarms.

Education

Nov 2025 – present
PhD
University of Trieste, Department of Engineering and Architecture
Supervisor: Prof. Erica Salvato

2022 – 2025
MSc in Electrical Energy and Control Systems Engineering
University of Trieste, Control Systems curriculum
Thesis: *Exploring Adaptive Reward and Perception Strategies in Multi-Agent Deep Reinforcement Learning for Intelligent Swarm Behavior*. Advisor: E. Salvato, co-advisor: E. Medvet.

2020 – 2022
BSc in Industrial Engineering
University of Trieste, Electrical Energy curriculum
Thesis: *Custom Inverter Design vs Commercial Solutions for Formula SAE Electric Powertrain*.

Research experience

Mar 2025 – present
Drone Laboratory, University of Trieste
With E. Salvato, F. Blanchini, G. Fenu, F. A. Pellegrino
Decentralized formation control for multi-drone systems: controllers implemented in Python, evaluated in Gazebo and validated on real drones in the laboratory arena, with MATLAB used for command execution.

Jan 2025 – Oct 2025
Research fellowship, Industrial Automation Laboratory, University of Trieste
Project "Applications of Artificial Intelligence Techniques to Control Systems", PI E. Salvato, with E. Medvet
Multi-agent reinforcement learning for coordination: MADDPG in continuous control under partial observability, studying how reward structure and observation range affect coordination and robustness.

Publications

2026
E. Salvato, A. Trevisan, F. Blanchini, G. Fenu, F. A. Pellegrino. **Optimal damping in decentralized formation control.** *Systems & Control Letters*, vol. 215, 106472. doi:10.1016/j.sysconle.2026.106472

Schools

Jul – Aug 2026
IEEE RAS Summer School on Multi-Robot Systems, Prague, Czech Republic

Awards

MSc
Best student of the MSc programme in Electrical Energy and Control Systems Engineering, first and second year

Other experience

2021 – 2023
Units Racing Team, Formula SAE, University of Trieste
Powertrain department: custom inverter design and comparison with commercial racing inverters, motor thermal models in Simulink.

Skills

Tools
Python, MATLAB/Simulink, Gazebo

Languages
Italian (native), English (B2)